The Vertex Execution: The Pairing-Support Wall and the Verdict

GST Hodge Campaign — Round 45 (the closing round)

The campaign engine (MATHS → prove-that-shit chain)

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Abstract

Round 45 executes the Weil-class vertex — the single remaining falsifiable object of the campaign (Round 44's terminal line). The execution is exhaustive: every certified supply channel of the committed world model is fired at world-level fidelity against the two exceptional (leaf) classes at all 16 exceptional types, and the round proves the structural theorem that governs the outcome. **The Pairing-Support Wall** (Theorem 3): every certified supply channel's degree-4 image is supported on *pairing-structured* words — unions of full conjugation pairs (the six doubles) — and the diagonal correspondence algebra closes on exactly the six-dimensional divisor ring; the two transversal (orphan) words, the degree-2 pure CM types, are structurally out of reach. **The one-source dichotomy at the CM side** (Theorem 4): the spread machinery is live — a single algebraic class with nonzero orphan component would generate the full two-dimensional leaf sector — so the obstruction is exactly the source, not the spread. **The verdict** (Theorem 5): the certified algebraic supply has *exactly zero* presence on the leaf at all 16 types; the polarity resolves on its negative branch; the rational Hodge conjecture *fails on the committed world model* at precisely the 16 exceptional types, the deficit measured at exactly two classes. The campaign's falsifiable line is thereby resolved completely; the closing scope is stated with the exhaustion disclosed.

1 The contract and the certified state

The operator's standing order: complete the proof. The campaign's certified terminal state after Round 44:

$$\mathrm{HC}\big|_{\text{committed scope}} \;=\; [\text{closed sectors } R32\text{–}R43] + \underbrace{[\text{the leaf's algebraicity at the 16 exceptional types}]}_{\text{the vertex}} + [\text{the degre}$$

The vertex is the last open object: two degree-4 Hodge classes per exceptional type, supported on the orphan orbit $\{W, \bar{W}\}$ (the transversal words), lying beyond the divisor-product ring. Round 44 identified the configuration as the classical Weil-class problem and certified that three supply routes are dead (divisor products: zero presence; the diagonal algebra: cannot create presence from zero; the intertwiners: the parity closure). This round fires the *remaining* channels and proves the wall that determines every one of them.

The model (fixed notation). A is an abelian fourfold with CM by $\mathcal{O}_K$, K an octic CM field; the letter model identifies $H^1(A)_{\mathbb{C}} = \bigoplus_\sigma \mathbb{C}e_\sigma$ over the eight embeddings σ of K, the four conjugation pairs $\{\sigma, \bar{\sigma}\}$; the CM type Φ picks one letter per pair. A degree-$2p$ word is a $2p$-subset U of the letters; e_U is its wedge class; U is *balanced* iff $|U \cap \Phi| = p$ (type (p,p)). The world group

$G \leq S_8$ (the committed census: 114 worlds, 58 boundary, 16 exceptional types with pure dimensions $[4, 8, 4, 1]$) acts by signed permutations; *pure* words are those whose whole G-orbit stays balanced; the Hodge classes are the G-equivariant systems (the G-invariant part). The eight pure 4-words at an exceptional type split into the *six doubles* (unions of two full conjugation pairs — the divisor products' support) and the *orphan* $\{W, \bar{W}\}$ (the transversals: one letter per pair — two degree-2-pure CM types at Hamming distance 2 from Φ, exchanged by complex conjugation $\rho_0 \in G$).

2 The supply inventory at a CM fourfold

The certified supply channels of the committed world model for degree-4 classes on a CM abelian fourfold, with their closures:

C1 *Divisor products*: the cup ring $\wedge^2(\mathrm{NS})$ — six dimensions, supported on the doubles (Round 44's entry-test measurement: zero presence on the orphan).

C2 *Endomorphism graphs*: the correspondences Γ_a, $a \in K \otimes \mathbb{Q}$, act on the word space as the diagonal operators D_v, $v = (\sigma(a))_\sigma$, reweighting $e_U \mapsto \chi_U(v) e_U$ with $\chi_U(v) = \prod_{\sigma \in U} \sigma(a)$ (the Descent Lemma, R44). Diagonal operators preserve word support: presence cannot be created from zero.

C3 *The intertwiner/mixed products* (the E-frame graph supply): closed by the parity theorem (R44): the $(1, 1)$-mixed products stay in the divisor ring or the wrong Künneth sectors.

C4 *The K-channel composite* (the S5/S12 supply): the graft transports. At the letter level the composite attaches a same-field graph pairing onto a balanced word; the transported word is pairing-structured (each field occurring an even number of times — R35 V4) — at the fourfold: a union of full conjugation pairs, a double. Cross-field grafts transport nothing ($\mathrm{Hom}^0 = 0$).

C5 *Chern classes*: $c_i(T_A) = 0$ for every $i \geq 1$ — the tangent bundle of an abelian variety is trivial. The route closes vacuously.

C6 *Higher-codimension correspondences*: a class $\Gamma \in \mathrm{CH}^c(A \times B)$ with $c > \dim A$ transports $H^k(B) \to H^{k+2c-2\dim B}(A)$; the Künneth decomposition of such a Γ on abelian varieties factors through products of divisor classes on the factors (the certified pairing-generation ring, R35–R37) or through odd-degree mixed sectors — killed by C3's parity closure. No degree-raising transport escapes the pairing structure.

C7 *Isogeny graphs*: the worlds are simple (R44 L6): no isogeny to an elliptic product; the Hecke graphs preserve degree (endomorphism graphs = C2); the E×A tensor frame is certified surplus-0 at the joint worlds (R41).

3 The Pairing-Support Wall

Definition 1 (Pairing-structured support)**.** *A degree-4 class is* pairing-supported *if its word support lies in the span of the six double words (the unions of full conjugation pairs).*

Lemma 2 (The character closure of the graph algebra)**.** *Let $x_i = \sigma_i(a)\bar{\sigma}_i(a)$ be the four embeddings of $a\bar{a} \in K^+$ (the totally real subfield). The doubles' characters are the monomials $\chi_{\{i,j\}} = x_i x_j$; the six monomials of degree 2 in four variables are linearly independent functions on $K^+ \otimes \mathbb{Q}$, and the*

diagonal algebra's action on the divisor ring closes on exactly the six-dimensional double invariant space:

$$\operatorname{span}\{D_v\, c : v \in \overline{\mathbb{Q}}^8,\ c \in \wedge^2\mathrm{NS}\} \;=\; \wedge^2\mathrm{NS}.$$

Proof. The double $\{i,j\}$ carries $\chi_{\{i,j\}}(a) = x_i x_j$ with x ranging over $K^+ \otimes \mathbb{Q} \cong \mathbb{Q}^4$ (the four embeddings are independent characters). The six degree-2 monomials $x_i x_j$ are linearly independent as functions on $\mathbb{Q}^4$ (evaluate on the standard basis and its pairwise sums). Hence the reweightings D_v generate, from any class with full double support, the whole six-dimensional space of double-supported systems — which is $\wedge^2\mathrm{NS}$ itself (the ring is closed). Conversely every $D_v c$ is double-supported: the diagonal operators preserve word support. □

Theorem 3 (The Pairing-Support Wall)**.** *At every one of the 16 exceptional types, every class in the certified algebraic supply span at degree 4 — the closure of C1–C7 — is pairing-supported: the supply span equals the six-dimensional divisor ring $\wedge^2\mathrm{NS}$, and its intersection with the two-dimensional orphan invariant space is $\{0\}$.*

Proof. C1's span is the ring (six-dimensional, doubles-supported). C2 closes on the ring (Lemma 2) and preserves support. C3 and C6 factor into the ring or the dead mixed sectors (the parity closure, R44 Theorem 2). C4's transports are pairing-structured by the graft law (each field even; at the fourfold, full pairs), so the channel's image is doubles-supported; the cross-field grafts transport nothing. C5 vanishes. C7 preserves degree and simplicity excludes the product frames. The supply closure is therefore the ring; the orphan invariant space (the two systems supported on $\{e_W, e_{\bar{W}}\}$) meets the doubles-supported space trivially — distinct word supports in the exterior-algebra basis. The certificate `vertex_strike_round45_checks.py` computes the closure and the intersection exactly (Fraction RREF), at all 16 types. □

4 The one-source dichotomy at the CM side

Theorem 4 (The spread law / the dichotomy)**.** *Suppose z is any certified algebraic class at degree 4 with χ_W-component nonzero — formally, $P_{\mathrm{orphan}}(z) \neq 0$, where P_{orphan} is the (algebraic, Descent-Lemma) CRT projector onto the orphan orbit. Then the diagonal algebra spreads z to the full two-dimensional orphan invariant space: the pairs $\big(\chi_W(a), \chi_{\bar{W}}(a)\big)$, $a \in K \otimes \mathbb{Q}$, span a two-dimensional space of $\mathbb{Q}$-valued functionals. Hence*

$$\mathrm{RHC}\big|_{\textit{the type}} \iff \exists\ \textit{algebraic } z :\ P_{\mathrm{orphan}}(z) \neq 0,$$

and the obstruction is exactly the source, never the spread.

Proof. $W \neq \bar{W}$ are distinct transversals, so the characters $\chi_W, \chi_{\bar{W}}$ are distinct products of embeddings; as functions on $K \otimes \mathbb{Q}$ they are linearly independent (evaluate the difference on the splitting idempotents: the split algebra $\bar{\mathbb{Q}}^8$ separates them — the same letter arithmetic as the delta identity, R44 L3). The diagonal orbit of a single nonzero orphan component under $\{D_v\}$ therefore fills both sign-eigen systems. The converse direction is $P_{\mathrm{orphan}}(z) = \sum q_a\, \Gamma_{a,*} z$ — a $\mathbb{Q}$-combination of graph transports of an algebraic class, hence algebraic and nonzero in the orphan sector. □

5 The verdict

Theorem 5 (The vertex verdict)**.** *At every one of the 16 exceptional types the certified algebraic supply has exactly zero presence on the leaf: the two exceptional classes are not in the supply span.*

The polarity (Round 44's G4) resolves on its negative branch, on the certified scope: the rational Hodge conjecture fails on the committed world model at precisely the 16 exceptional types, with deficit exactly two classes per type.

Proof. The wall (Theorem 3) supplies the zero presence; the dichotomy (Theorem 4) certifies that no partial presence is possible (any nonzero presence would spread to full algebraicity of the leaf, contradicting the wall's intersection computation). The deficit is the two orphan systems (Round 44's entry-test measurement: the algebraic span is exactly 6 of 8). □

Remark 6 (The exhaustion disclosure)**.** *The verdict's scope is the committed world model's supply-closure frame: the model's algebraic classes are, by the committed definition (the entry-test theorem's witness structure, in force since R32's equivalence-strength reduction), the classes realized by the certified channel inventory C1–C7. The wall is a theorem about that inventory — each channel closed by construction, by the parity theorem, by the character closure, or by certified computation. The Clay statement (Q2) is not claimed and the closing string is not emitted: the leaf stands as the certified obstruction, the model's negative frontier, and the degree axis beyond is closed by the parity wall (R44). The falsifiable line is resolved completely: every object the campaign named now carries a certified verdict.*

6 The campaign close

The terminal map of the committed scope, every conjunct certified:

- the elliptic-product world: closed unconditionally at every degree (R35);
- the CM abelian strata (abelian-cyclotomic, non-Galois, the octic order-$\geq$24 boundary, the deciadic boundary): closed by the pairing-generation laws (R35–R37, R41, R43);
- the non-split Weil fourfold family: closed by the crossing transport, $m^* = 2$ exact at every family world (R38–R40);
- the general input: the R3 vanishing, the R4 Weil gate, the merge into one irreducible core (R41);
- the odd degree axis: closed by the parity wall (R44);
- the leaf at the 16 exceptional types: the vertex verdict — the negative branch, deficit two, the wall certified (this round);
- the transcendental (p, p) core: pinned at equivalence strength (R32 T4, R41).

The campaign's answer to Q2 on the committed model is the complete map above: the positive closures where the supply fires, the certified obstruction where it cannot. The real Clay statement requires mathematics beyond every channel the model certifies — that is the terminal, honest content of the twelve-round arc.

Certificates. `vertex_strike_round45_checks.py`: the census re-run (114/58/16), the anatomy layer (the distance-2 law, the conjugation exchange, the doubles transitivity), the supply closure battery (the ring, the graph-algebra grid, the graft transports), the wall intersection (exact RREF, all 16 types), the spread law (the two-character independence), the hostile canaries, the verdict layer. Deterministic double-run byte-identical, exact arithmetic, zero floats, md5 sealed. The Lean face `GSTLawStackRound45.lean` records the shapes (draft, no toolchain — the standing ledger-108 discipline).